Machine Learning for Customer Churn Prediction

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1 Summary

This work formulates the customer churn retention problem as a Markov Decision Process (MDP). The objective is to train a Reinforcement Learning (RL) agent to select the best retention strategy for a customer with a probability of churn. Although the available dataset does not contain labeled data describing the outcome of different retention strategies, it provides sufficient customer and service information to establish an environment where the agent can learn which actions can reduce the probability of churn.

2 MDP Formulation

The MDP is composed of a state space containing customer features, an action set representing different retention strategies, stochastic transitions based on the customer's acceptance of an offer, and a reward function based on the reduction of the predicted churn probability and the cost of the selected action. A Deep Q-Network (DQN) algorithm is used as the RL agent to determine the action with the highest expected reward.

2.1 Problem Statement

The objective of the MDP is to determine the best retention strategy for a customer according to their characteristics and current probability of churn. The RL agent receives the customer's state and selects an action from a set of possible retention strategies. The selected action can be accepted or rejected by the customer, introducing stochasticity into the environment.

The customer's churn probability is obtained using a supervised learning model. This probability is used by the RL environment to evaluate whether the selected retention strategy improves the customer's situation. The objective of the agent is therefore to reduce the probability of churn while considering the cost associated with some retention strategies.

2.2 State Space

The state consists of the features that describe the customer and their current service configuration. The original Telco Customer Churn dataset contains 19 customer and service features, without count the customer id. In addition, 2 features are created through feature engineering, and 1 additional feature is generated by the supervised learning model. This feature represents the predicted probability of the customer being churn.

For this reason, each state is represented as a vector of 22 elements:

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State = [19 original features, 2 engineered features, churn probability]
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2.3 Action Set

The action set contains 15 possible retention strategies. Each action modifies specific features of the customer's state. The actions are divided into three categories: contract plan changes, monthly charge discounts, and new service offers.

2.3.1 Change the Contract Plan

Customers with a month-to-month contract usually have a higher probability of churn than customers with one-year or two-year contracts. Therefore, the agent can offer a change in the customer's contract duration. These actions have zero cost.

1. Offer change to a One Year contract.
2. Offer change to a Two Years contract.

2.3.2 Discount in the Monthly Charge

High monthly charges can increase the probability of customer churn. For this reason, the agent can offer a discount of 5%, 10%, or 15% on the customer's monthly charge.

These actions generate a cost for the company. The cost is proportional to the size of the discount and is defined as the discount percentage multiplied by one hundred.

3. Offer a 5% discount on the Monthly Charges.
4. Offer a 10% discount on the Monthly Charges.
5. Offer a 15% discount on the Monthly Charges.

2.3.3 Offer a New Service

The company provides 10 services that can be offered to customers. When a service is accepted, it is added to the customer's plan and its estimated price is added to the Monthly Charges. The service prices were estimated from the dataset to obtain approximate values in USD.

There are two restrictions for these actions. Multiple Lines cannot be offered if the customer does not have Phone Service. The other services cannot be offered if the customer does not have Internet Service. These actions have zero cost.

6. Offer Phone Service (USD 20).
7. Offer Multiple Lines (USD 5).
8. Offer Internet Service DSL (USD 25).
9. Offer Internet Service Fiber Optic (USD 50).
10. Offer Online Security (USD 5).
11. Offer Online Backup (USD 5).
12. Offer Device Protection (USD 5).
13. Offer Tech Support (USD 5).
14. Offer Streaming TV (USD 10).
15. Offer Streaming Movies (USD 10).

2.4 Transition Dynamics

The selected action directly affects the customer's state. However, before modifying the state, the environment determines whether the customer accepts or rejects the offered strategy. This makes the transition stochastic because the same action can produce different outcomes.

The acceptance probabilities are defined according to the type of retention strategy:

- Change the contract plan: 85% acceptance and 15% rejection.
- Discount in the monthly charge: 95% acceptance and 5% rejection.
- Offer a new service: 90% acceptance and 10% rejection.

The transition function can be represented as:

$$s_{t+1} = f(s_t, a_t) \tag{1}$$

The acceptance or rejection of an action is stochastic. However, once an action is accepted, the modification applied to the customer's state is deterministic. Therefore, the same state and the same accepted action produce the same resulting state.

2.5 Reward Function

The reward function measures the benefit obtained from applying a retention strategy. It is based on the difference between the customer's churn probability before and after the action, considering the cost associated with the selected strategy.

The reward is calculated using the following function:

$$Reward = (P_{before} - P_{after})(1 - Cost) \tag{2}$$

- `P_before`: Churn probability predicted by the supervised model at the beginning of the episode.
- `P_after`: Churn probability predicted by the supervised model after the action is accepted and the customer's state is modified.
- `Cost`: Cost associated with the selected retention strategy.
- `Reward`: Benefit obtained from reducing the customer's probability of churn while considering the cost of the action.

A reduction in the predicted churn probability produces a positive reward, while an increase in the probability or a high action cost reduces the resulting reward.

2.6 Episode

An episode starts with a customer from the Telco Customer Churn dataset. The customer's initial state is provided to the supervised learning model, which predicts the probability of churn. This information is then provided to the RL agent.

The agent uses the DQN algorithm to select the retention strategy with the highest expected reward. The selected action is then evaluated by the environment, which determines whether the customer accepts or rejects the offer according to the corresponding acceptance probability.

If the customer accepts the offer, the action modifies the customer's state. The supervised model then predicts the new probability of churn, which is compared with the probability before the action to calculate the reward. The episode terminates after this reward is obtained.

If the customer rejects the offer, the customer's state remains unchanged and the reward is zero. The episode then terminates.

3 Environment

4 Agent algorithm

5 Hyperparameter

6 Sample trajectory visualizations

7 Discussion

8 Conclusion

References

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